INCH-POUND
MIL-P-18080A(SHIPS)
AMENDMENT 2
7 April 1989
SUPERSEDING
AMENDMENT 1
10 January 1967

MILITARY SPECIFICATION

PLASTIC SHEETS, VINYL, FLEXIBLE, TRANSPARENT,
OPTICAL QUALITY

This amendment forms a part of MIL-P-18080A(SHIPS), dated 2 March 1956, and is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* 2.1, MIL-T-19102: Delete "Gas" and substitute "Protective".

2.1: Delete:

"NAVY DEPARTMENT
General Specifications for Inspection of Material."

* 2.1, MIL-STD-105: Delete "Inspection Table" and substitute "Procedures and Tables".

PAGE 2

* 2.2: Delete and substitute:

"2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:57Z
Check the source to verify that this is the current version before use.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 412 - Standard Test Methods for Rubber Properties in Tension. (DoD adopted)

D 746 - Standard Test Method for Brittleness Temperature of Plastics and Elastomers by Impact. (DoD adopted)

D 792 - Standard Test Methods for Specific Gravity (Relative Density) and Density of Plastics by Displacement. (DoD adopted)

D 1003 - Standard Test Method for Haze and Luminous Transmittance of Transparent Plastics. (DoD adopted)

D 3951 - Standard Practice for Commercial Packaging. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)"

* Add as new paragraph 2.3:

"2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained."

PAGE 4

4.1 through 4.3: Delete and substitute:

* "4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:57Z
Check the source to verify that this is the current version before use.

submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Sampling for quality conformance inspection. Sampling for quality conformance inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated. For purposes of sampling, an inspection lot for inspection and tests shall consist of all material of the same type, machine run and thickness range submitted for delivery at one time.

4.2.1 Inspection of materials and components. In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification. In the event of conflict, this specification shall govern.

4.2.2 Inspection of the end item.

4.2.2.1 Examination of the end item. Examination of the end item shall be made in accordance with the classification of defects, inspection levels and acceptable quality levels (AQLs) set forth below. The lot size, for purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of plastic sheets for examination in 4.2.2.1.1, 4.2.2.1.2 and in units of shipping containers for examination in 4.2.2.1.3.

4.2.2.1.1 Examination of the end item for defects in appearance and workmanship. The sample unit for this examination shall be one sheet of the specified type plastic material.

Examine	Defects
Appearance and workmanship	Not uniform in transparency.
	Any blisters, bubbles, wrinkles, creases, cuts, tears, holes, dents, heat marks, or graininess which will affect serviceability.
	Any scratches, dirt, imbedded particles or other defects which will reduce visibility or affect appearance.
Finish	Not finished to a polished or smooth finish.
Color	Not colorless and transparent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:57Z
Check the source to verify that this is the current version before use.

4.2.2.1.2 Examination of the end item for dimensional defects. The sample unit for this examination shall be one sheet of the specified type plastic material.

Examine	Defects
Length and width	Varies by more than minus 0 plus 1/4 inch from size specified.
Thickness	Varies by more than plus or minus the tolerance specified (see table II).

4.2.2.1.3 Examination of the preparation for delivery. An examination shall be made to determine that packaging, packing and markings comply with the requirements of section 5. The sample unit for this examination shall be one shipping container, fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects.

Examine	Defects
Packaging	Not level specified, not in accordance with contract requirements.
	Packaging material not as specified.
	Paper interleaves missing, not extending over full area between sheets, or not as specified.
Packing	Not level specified, not in accordance with contract requirements.
	Any nonconforming component, component missing, damaged or otherwise defective, affecting serviceability.
	Container not as specified, closures not accomplished by specified or required methods or materials.
	Inadequate application of components such as: incomplete closure of case liners, container flaps, loose or inadequate strapping, bulged or distorted containers.
Count	Less than specified or indicated quantity of sheets per container.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:57Z
Check the source to verify that this is the current version before use.

Examine	Defects
Weight	Gross or net weight exceeds specified requirements.
Markings	Exterior markings omitted, illegible, incorrect, incomplete or not in accordance with contract requirements.
	Precautionary markings missing or not as specified.

4.2.3 Inspection levels and acceptable quality levels (AQLs) for examination. The inspection levels for determining the sample size and the acceptable quality levels (AQLs) expressed in defects per 100 units, shall be as follows:

Examination paragraph	Inspection level	AQL
4.2.2.1.1	I	2.5
4.2.2.1.2	S-3	2.5
4.2.2.1.3	S-2	4.0

4.3 Testing of the end item. The end item shall be tested for the applicable characteristics as indicated in table III, on each lot presented for examination. The sample unit shall be one sheet. The inspection level for determining the sample size shall be S-1, except that no less than three sample units shall be randomly selected throughout the lot. The acceptable quality level (AQL) shall be 6.5 expressed in defects per 100 units.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:57Z
Check the source to verify that this is the current version before use.

TABLE III. Instructions for testing.

Characteristic	Specification reference		Requirements applicable to individual lot		Number determinations per unit	Pass or fail	Numerically to nearest 1/
	Requirement	Test method	unit	average			
Physical properties	3.2 and Table I						
Specific gravity	Table I	4.5.1	X	-	Average of 5	-	.01
Tensile strength -							
lengthwise	Table I	4.5.2	X	-	Average of 5	-	psi
crosswise		4.5.2	X	-	Average of 5	-	psi
Elongation -							
lengthwise	Table I	4.5.3	X	-	Average of 5	-	percent
crosswise		4.5.3	X	-	Average of 5	-	percent
Brittleness temperature	Table I	4.5.4	X	-	2	-	degree(°)C.
Light transmission A	Table I	4.5.5	X	-	Average of 5	-	percent
Light transmission B		4.5.5	X	-	Average of 5	-	percent
Haze	Table I	4.5.6	X	-	Average of 5	-	0.1 percent

1/ Test results shall include all values on which results are based."

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:57Z
Check the source to verify that this is the current version before use.

PAGE 6

4.6: Delete.

* 5: Delete and substitute:

"5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)"

* 5.1: Delete and substitute:

"Preservation. Preservation shall be level A or Commercial, as specified (see 6.2)."

* 5.1.2: Delete and substitute:

"5.1.2 Commercial. Sheets of plastic material shall be slip-sheeted or interleaved in accordance with ASTM D 3951."

PAGE 7

* 5.2: Delete and substitute:

"5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2)."

* 5.2.3: Delete and substitute:

"5.2.3 Commercial. Plastic sheets preserved as specified in 5.1 shall be packed in accordance with ASTM D 3951."

* 5.3, line 2: Delete all after "order," and substitute: "level A and B exterior shipping containers shall be marked in accordance with MIL-STD-129, commercial exterior containers in accordance with ASTM D 3951, and shall include barcode markings on all exterior containers."

* 6.2: Delete and substitute:

"(c) Level or commercial preservation and packing required (see 5.1 and 5.2)."

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:57Z
Check the source to verify that this is the current version before use.

Patent notice: Delete.

Custodian: Delete.

The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Preparing activity:
Navy - SH
(Project 9330-NB16)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:57Z
Check the source to verify that this is the current version before use.